Research Memo: Neural Network Activation Functions

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Research Question: When using open-ended descriptions to predict dominant transportation mode, which hidden-layer activation function achieves the highest validation macro-F1 score?

Answer, Response, + Summary of Results

To answer this research question, I tested four different activation functions in the neural network's hidden layer while holding the architecture, optimizer (Adam), and training parameters constant. The activation functions tested were ReLU, tanh, Leaky ReLU, and GELU.

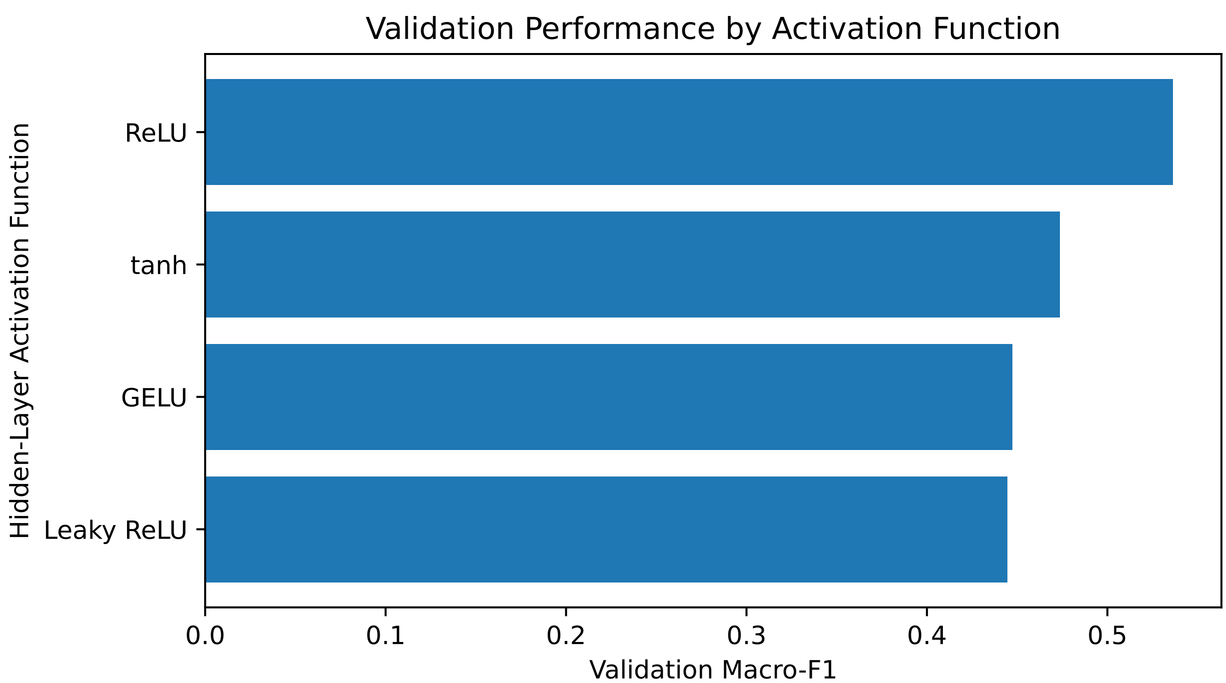

Figure 1: Validation macro-F1 scores for the four activation functions.

As shown in Figure 1, ReLU achieved the highest validation macro-F1 score of 0.536, followed by tanh with a score of 0.474.

Table 1: Validation performance for each activation function.

Activation Function	Validation Macro-F1	Validation Accuracy	Epochs Trained
ReLU	0.536	0.667	50
tanh	0.474	0.583	41
GELU	0.447	0.542	47
Leaky ReLU	0.444	0.542	45

The complete performance comparison is shown in Table 1. These results indicate that ReLU is the most effective choice among the four activation functions tested for this neural network architecture when predicting dominant transportation mode from open-ended text responses.

The full model-training code is available in the project repository:

https://github.com/emilyhuffaker/DAGE/blob/main/scripts/activation_functions_test.py

What Questions or Uncertainties Remain?

Would these activation function rankings remain consistent with different architectures or hyperparameters?

Would the results remain stable across different training, validation, and test splits?

How would these activation functions perform on the apprehension prediction task?